

Answer **all** the questions

Question 1 [Total: 10 marks]

Phospholipids are components of cell membranes. Fig. 1.1 below shows a schematic representation of a phospholipid molecule.

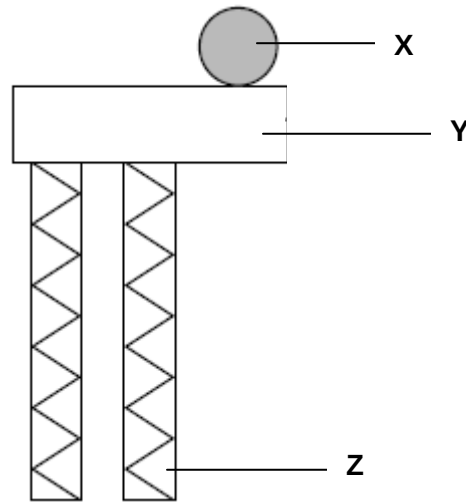


Fig. 1.1

- (a) Label the parts **X**, **Y** and **Z**.

X _____
Y _____
Z _____ [3]

- (b) State how a phospholipid molecule differs in structure from a triglyceride molecule.

 _____ [2]

The lipid bilayer of many cell membranes is not composed exclusively of phospholipids. It often also contains cholesterol, glycolipids and proteins.

- (c) Explain the functions of the following in cell membranes

- (i) cholesterol

 _____ [1]

(ii) glycolipids

[1]

Fig. 1.2 below shows a schematic representation of two types of transport proteins found in cell membranes.

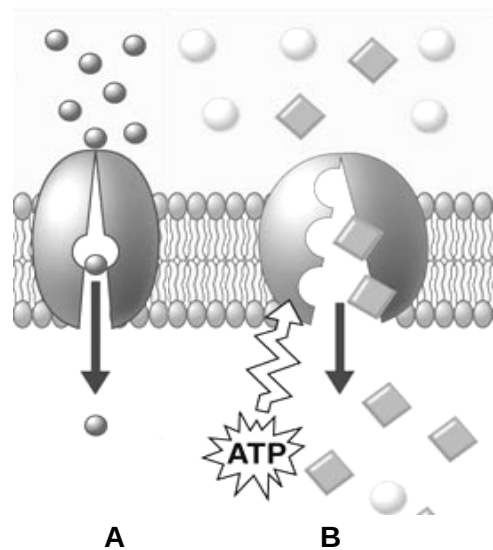


Fig. 1.2

(d) State **one** difference between transport proteins A and B.

[1]

(e) Explain why transport proteins are required for movement of substances across membranes.

[2]

Question 2 [Total: 10 marks]

The common fruit fly, *Drosophila melanogaster*, found in Australia is an economic pest. In a population of flies studied, there are flies with black body and normal wings. There are also flies with white body and vestigial wings.

The allele for black body was assumed to be dominant to the allele for white body (absence of pigment). The allele for normal wings was also assumed to be dominant to the allele for vestigial wings.

- (a) Assuming the alleles of the two genes segregate independently, state the offspring phenotypes and the respective phenotypic ratio of a cross between two heterozygous flies.

[2]

When an actual cross between a homozygous black body fly with normal wings and a homozygous white body fly with vestigial wings was carried out, all offspring have grey body and normal wings.

- (b) It was stated that the allele for black body is dominant to the allele for white body. State if this assumption is valid.

[1]

- (c) Explain the biochemical basis for the presence of grey body flies.

[3]

In addition, when the grey body normal wing offspring is mated with an individual with white body and vestigial wings, the following results are obtained.

grey body and normal wings	167
grey body and vestigial wings	150
white body and normal wings	161
white body and vestigial wings	156

Use the following symbols:

B represents the allele for black body

b represents the allele for white body

N represents the allele for normal wings

n represents the allele for vestigial wings

(d) Use a genetic diagram to explain the cross above.

Question 3 [10 marks]

Fig. 3.1 is an electron micrograph of a eukaryote interphase nucleus.

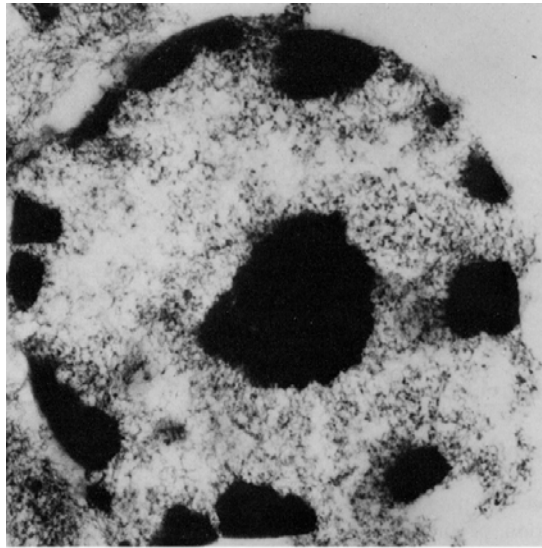


Fig. 3.1

- (a) Describe the structural organization of DNA in the eukaryote interphase nucleus.

[4]

In eukaryotic transcription apart from structural changes, control elements also play a major role in controlling gene expression at the transcriptional level.

- (b) Explain the role of **one** control element that is involved in influencing eukaryotic transcription.

[2]

Fig. 3.2 shows a protein which plays a central role in eukaryotic transcription. The binding of this protein forms the first step in the formation of the transcription initiation complex.

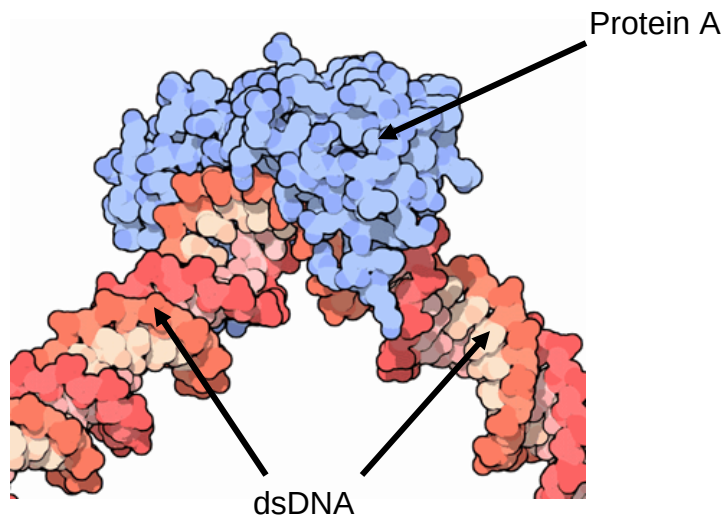


Fig. 3.2

- (c) Based on Fig. 3.2, precisely identify Protein A and the region on the DNA that Protein A binds.

[2]

- (d) Suggest how Protein A associates with the DNA.

[2]

Question 4 [Total: 10 marks]

A recombinant plasmid can be formed when the plasmid is cut and ligated with a DNA fragment containing the gene of interest. Fig. 4.1 shows a plasmid vector, pBR322, and the DNA fragment containing the gene of interest (β -gal).

β -gal codes for the enzyme β -galactosidase, which catalyses the following reaction:

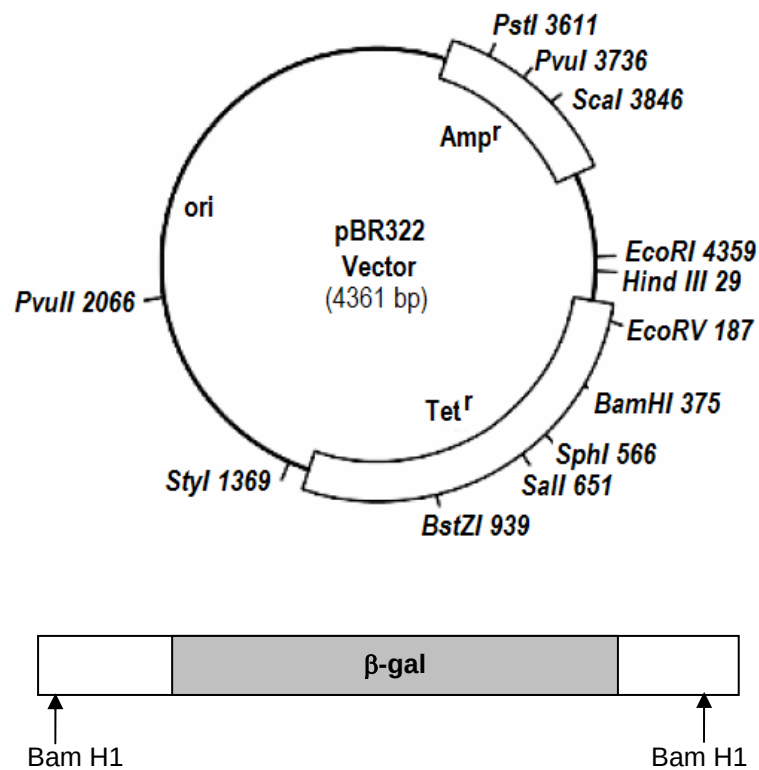
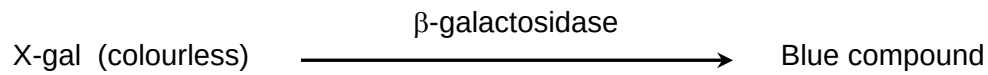


Fig. 4.1

- (a) With reference to Fig. 4.1, draw a diagram of the resulting recombinant plasmid in the space below, showing **all** the genes present and **ONLY** the relevant restriction site.

[2]

The recombinant plasmid was inserted into the bacteria, *Escherichia coli*, which did not previously exhibit any antibiotic resistance.

- (b) Explain whether transformed *E. coli* which contains the recombinant plasmid will be able to survive on a growth medium containing tetracycline.

[2]

The *E. coli* was then cultured on a growth medium containing X-gal and three different antibiotic tablets. Fig. 4.2 shows the results.

Key:

Tet – tetracycline tablet

Kan – kanamycin tablet

Amp – ampicillin tablet

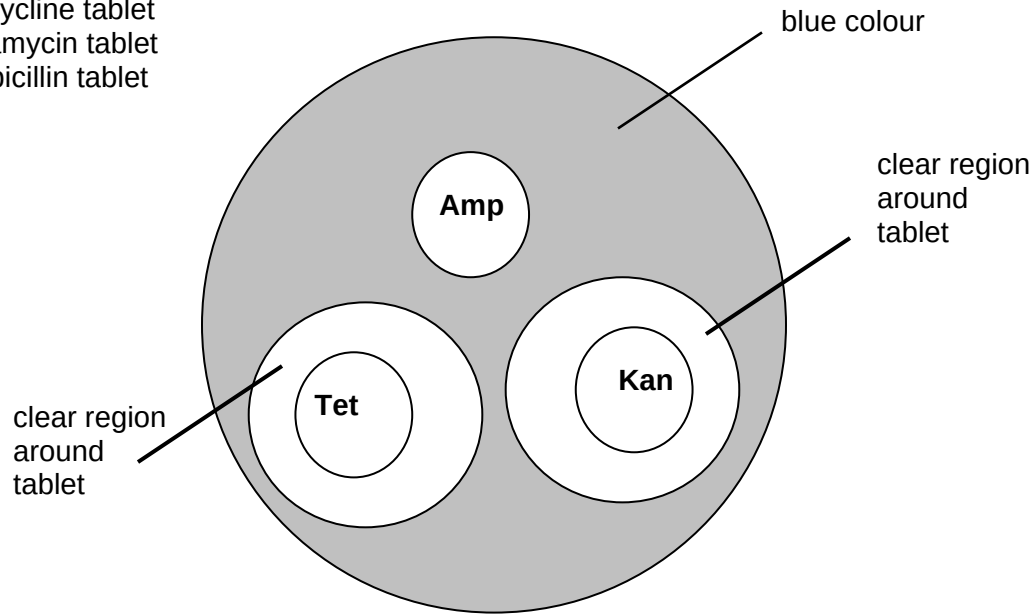


Fig. 4.2

- (c) Explain the presence of the clear region around the kanamycin tablet.

[1]

- (d) Explain the presence of the blue regions around the ampicillin tablet.

[2]

- (e) Explain why replica plating is **not** necessary for the selection of the transformed *E. coli*.

[3]

End of Section A

Section B

Answer **one** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a) Outline the main differences between a fibrous protein and DNA. [6]
- (b) Describe the similarities between oxidative phosphorylation and non-cyclic photophosphorylation. [6]
- (c) Describe the use of restriction enzymes in genetic engineering. [8]
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- 6 (a) Compare the process of photosynthesis and aerobic respiration. [8]
- (b) Distinguish between competitive and non-competitive inhibition of enzymes. [7]
- (c) Using a named example, explain the advantages of being able to transfer human genes into bacteria. [5]

End of Paper